

Generative Environments Maker for Self-Evolved Agentic RL

Yang Liu^{*}, Shuhan Qin^{*}, Jun Bai^{*}, Jiaqi Li^{*}, Xiaobo Wang^{*}
Tong Wu, Zhe Li, Yanting Wang, Zixia Jia, and Zilong Zheng[✉]

NLCo Lab, Beijing Institute for General Artificial Intelligence

zlzheng@bigai.ai



Code: <https://github.com/bigai-nlco/AgentGEM>



Models: <https://huggingface.co/AgentGEM>



Website: <https://bigai-nlco.github.io/AgentGEM>



W&B Logs: <https://wandb.ai/agent-gem/FusedRL>

1. Introduction

Our innovations are summarized as three folds:

- **Environments & Tasks Generation:**
- **Environmental Trajectory Discovery:**
- **Simple yet Effective Mixed RL Recipe:** Baseline advantage estimation like GRPO with simple strategies of partial credit assignments.

2. Preliminaries

2.1. Preliminary I: Reinforcement Learning Environments

RL Environments. RL environments are the playgrounds where agents learn: at each timestep, the environment provides a state $s_t \in S$, the agent takes an action $a_t \in A$ under policy π_θ , and the environment returns a reward $r_t = R(s_t, a_t)$ and transitions to $s_{t+1} \sim T(\cdot | s_t, a_t)$. This state-action-reward loop repeats until a terminal condition is reached, defining an episode over which the agent’s cumulative reward is evaluated; the agent adapts its policy across episodes to maximize this return (Sutton et al., 1999). Recently, RL environments have been extended to language agents use cases, for example, simulating a web browser (Zhou et al., 2024) or enforcing formatting rules for a conversational scenario (Barres et al., 2025).

POMDP Formulation. We cast the problem within the Partially Observable Markov Decision Process (POMDP) framework (Åström, 1965; Smallwood and Sondik, 1973). Formally, a POMDP is characterized by the tuple (S, A, T, R) . The state space S typically encompasses the interaction history, search results, and retrieved webpage content; since the agent only perceives this history rather than any latent environment state, S effectively serves as the observation space under partial observability. The action space A consists of tokens generated by the agent, from which tool-calling instructions can be parsed via designated tags, such as `<tool_call><function=web_search>...</function></tool_call>` (Jin

^{*} Core Contributors.

✉ Corresponding author(s): Zilong Zheng.

et al., 2025). The transition function $T(s' | s, a)$ governs how the environment evolves in response to the agent’s actions, e.g., appending retrieved content to the history. At each timestep t , the agent observes a state s_t and, following its policy $\pi_\theta : S \rightarrow A$, produces an action a_t , receiving reward $R(s_t, a_t)$. The agent’s objective is to learn a policy that maximizes the expected cumulative return:

$$\mathcal{J}(\pi_\theta) = \mathbb{E} \left[\sum_{t=0}^{\infty} R(s_t, a_t) \mid a_t \sim \pi(s_t) \right]. \quad (1)$$

2.2. Preliminary II: Agentic Reinforcement Learning

To train a language agent, modern policy gradient RL algorithms like PPO (Schulman et al., 2017) and GRPO algorithm (Shao et al., 2024) are widely adopted. For each input question x , we sample a group of G trajectories $\tau_1, \tau_2, \dots, \tau_G$, where each trajectory takes the form $\tau_i = (s_0^i, a_0^i, s_1^i, \dots, s_{T_i}^i)$. The agent is then optimized using the following objective: Here, ϵ is a hyperparameter controlling the clipping range, while $\hat{A}_i$ denotes the advantage assigned to the i -th trajectory, derived from the relative rewards observed across all trajectories within the same group.

$$\mathcal{J}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, \{\tau_i\}_{i=1}^G \sim \pi_{\theta_{old}}(\cdot | x)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{\sum_{t=0}^{T_i-1} |a_t^i|} \sum_{t=0}^{T_i-1} \sum_{j=1}^{|a_t^i|} \min \left(\frac{\pi_\theta(a_{t,j}^i | s_t, a_{t,<j}^i)}{\pi_{\theta_{old}}(a_{t,j}^i | s_t, a_{t,<j}^i)} \hat{A}_i, \text{clip} \left(\frac{\pi_\theta(a_{t,j}^i | s_t, a_{t,<j}^i)}{\pi_{\theta_{old}}(a_{t,j}^i | s_t, a_{t,<j}^i)}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_i \right) \right], \quad (2)$$

Accompanied by Agentic RL, the online rejection sampling methods are often adopted to enhance the training effectiveness, such as the dynamic filtering method by DAPO (Yu et al., 2025). Specifically, dynamic filtering remove task groups where all episodes yield identical returns (resulting in zero advantages). Each sampled trajectory is evaluated using an indicator $\mathbb{1}_{\text{correct}}(\tau)$, applying this filter yields the non-zero advantage training batch of valid task groups: $\mathcal{B}_{\text{valid}} = \{(x_i, \tau_j^i) \mid i \leq N, j \leq K, \text{ s.t. } \tau_j^i \sim \pi_\theta(\cdot | x_i), \mathbb{1}_{\text{correct}}(\tau_j^i) = 1\}$ for later actor update.

3. Related Work

Scaling RL Environments.

Self-Evolved Agents.

Limitations of Agentic RL on Sample Efficiency.

4. GEM

4.1. Schema Definition & Design

4.2. Environment Data Generation

Table 1: Statistics of synthesized environments and tasks generated by AgentGEM.

# Environments	# Tasks	# Trajectories	Task Format	Tool Registry	Reasoning Depth	Verification
MCP						
Web Search						
SWE						
Terminal						

Table 2: Comparison with all baselines on eight benchmarks. AgentGEM outperforms most of other methods. Average is the macro mean across all samples of tasks.

Method	Tool Use		Conversation		Scientific QA		SWE / CLI		Avg. Results
	BFCL-v3	ACEBench	τ^2 -Bench	VitaBench	GPQA	MedQA	SWE-bench	TB-2.0	
ENVIRONMENT SCALING									
Toucan (?)									
Simia-Tau (?)									
RLVE (?)									
AWM (?)									
EnvScaler (?)									
IMITATION LEARNING									
SFT w/o Trajectory (Wei et al., 2022)									
SFT w/ Trajectory (?)									
ADVANCED RLVRs									
GRPO (Shao et al., 2024)									
StarPO (Wang et al., 2025)									
ARPO (?)									
STATIC CURRICULUM									
Data-balance RL (?)									
Easy-to-hard RL (?)									
ADARFT (Shi et al., 2025)									
AGENTGEM (Ours)									
AGENTGEM-4B									
AGENTGEM-8B									

4.3. Agent Trajectory Discovery and Policy Self-Distillation

4.4. Agentic RL in Synthesized Environments

5. Empirical Results

5.1. Experimental Setup

Benchmarks. We assess the generalization of agents on unseen tasks using three types of benchmarks including (1) General Tool Use: VitaBench (He et al., 2025), τ^2 -Bench (Barres et al., 2025), and BFCL (v3) (Patil et al., 2025); (2) Web Search QA: HLE (Scale et al., 2026), BrowseComp-Plus (Chen et al., 2026), MedQA (Jin et al., 2021), GPQA (Rein et al., 2024), and Bamboogle (Press et al., 2023); and (3) Software Engineering and Command-line Problems: SWE-Bench Verified (Jimenez et al., 2024) and Terminal-Bench 2.0 (Merrill et al., 2026).

Compared Baselines.

Evaluation Metrics. All tasks in the work are evaluated using the algebraic mean of hard exact match, which is also equivalent to pass@1 accuracy under strict extraction and comparison.

Implementation Details. We use Qwen3-4B-Thinking-2507 (?) as the base model and employ veRL (Sheng et al., 2024) and rLLM (?) for agentic RL post-training. We set train and mini batch sizes to 64 for strict on-policy updates, and a rollout size of 16 per training prompt. **Inference Setup:** We employ random sampling (temperature $\tau = 0.6$ and $top-p = 0.95$ with a maximum output length of 2,048. For behavioral cloning and RLVR baselines, we use Qwen3-8B-Base for full-parameter SFT or online RL. We perform five runs per test set and calculate the mean and standard deviation of the performance. We also report one-sided p-values for statistical significance.

5.2. AGENTGEM Improves RLVR Performance and Efficiency

Out-of-Distribution Performance.

5.3. Analyses and Discussion

Advantages of AGENTGEM

Impact of Task Mixing Recipe

Investigation on Training Sample Efficiency

Case Studies: AGENTGEM Improves the Quality of the Real-World Desiderata.

6. Conclusion

We introduce AGENTGEM, a novel training framework tailored to enhance both effectiveness and efficiency of multi-domain training for RLVR. AGENTGEM effectively harmonizes diverse rule-based reasoning capabilities across various reasoning tasks, resulting in demonstrably higher in-distribution and out-of-distribution performance efficiently compared to prior rule-based reasoners, state-of-the-art LRMs, strong supervised training approaches, and RLVR methods. We aim to explore further research into data-centric approaches to facilitate reasoning model efficiency in future work.

Limitations & Future Work

Acknowledgments

References

- Richard S Sutton, David McAllester, Satinder Singh, and Yishay Mansour. Policy gradient methods for reinforcement learning with function approximation. In S. Solla, T. Leen, and K. Müller, editors, *Advances in Neural Information Processing Systems*, volume 12. MIT Press, 1999. URL https://proceedings.neurips.cc/paper_files/paper/1999/file/464d828b85b0bed98e80ade0a5c43b0f-Paper.pdf.
- Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. Webarena: A realistic web environment for building autonomous agents. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=oKn9c6ytLx>.
- Victor Barres, Honghua Dong, Soham Ray, Xujie Si, and Karthik Narasimhan. τ^2 -bench: Evaluating conversational agents in a dual-control environment. *arXiv preprint arXiv:2506.07982*, 2025.
- Karl Johan Åström. Optimal control of markov processes with incomplete state information i. *Journal of mathematical analysis and applications*, 10:174–205, 1965.
- Richard D Smallwood and Edward J Sondik. The optimal control of partially observable markov processes over a finite horizon. *Operations research*, 21(5):1071–1088, 1973.
- Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Sercan O Arik, Dong Wang, Hamed Zamani, and Jiawei Han. Search-r1: Training LLMs to reason and leverage search engines with reinforcement learning. In *Second Conference on Language Modeling*, 2025. URL <https://openreview.net/forum?id=Rwhi91ideu>.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Y Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Qiyang Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Tiantian Fan, Gaohong Liu, Lingjun Liu, Xin Liu, et al. Dapo: An open-source llm reinforcement learning system at scale. *arXiv preprint arXiv:2503.14476*, 2025.
- Wei He, Yueqing Sun, Hongyan Hao, Xueyuan Hao, Zhikang Xia, Qi Gu, Chengcheng Han, Dengchang Zhao, Hui Su, Kefeng Zhang, et al. Vitabench: Benchmarking llm agents with versatile interactive tasks in real-world applications. *arXiv preprint arXiv:2509.26490*, 2025.

- Shishir G Patil, Huanzhi Mao, Fanjia Yan, Charlie Cheng-Jie Ji, Vishnu Suresh, Ion Stoica, and Joseph E. Gonzalez. The berkeley function calling leaderboard (BFCL): From tool use to agentic evaluation of large language models. In *Forty-second International Conference on Machine Learning*, 2025. URL <https://openreview.net/forum?id=2GmDdhBdDk>.
- AI Scale, Paolo Giordano, HLE Contributors Consortium, et al. A benchmark of expert-level academic questions to assess ai capabilities. *Nature*, 649:1139–1146, 2026.
- Zijian Chen, Xueguang Ma, Shengyao Zhuang, Ping Nie, Kai Zou, Sahel Sharifymoghaddam, Andrew Liu, Joshua Green, Kshama Patel, Ruoxi Meng, Mingyi Su, Yanxi Li, Haoran Hong, Xinyu Shi, Xuye Liu, Hosna Oyarhoseini, Nandan Thakur, Crystina Zhang, Luyu Gao, Wenhu Chen, and Jimmy Lin. BrowseComp-plus: A fair and disentangled evaluation benchmark for deep search agents. In Maria Liakata, Viviane P. Moreira, Jiajun Zhang, and David Jurgens, editors, *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 22349–22370, San Diego, California, United States, July 2026. Association for Computational Linguistics. ISBN 979-8-89176-390-6. doi: 10.18653/v1/2026.acl-long.1023. URL <https://aclanthology.org/2026.acl-long.1023/>.
- Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14):6421, 2021.
- David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPQA: A graduate-level google-proof q&a benchmark. In *First Conference on Language Modeling*, 2024. URL <https://openreview.net/forum?id=Ti67584b98>.
- Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah Smith, and Mike Lewis. Measuring and narrowing the compositionality gap in language models. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5687–5711, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.378. URL <https://aclanthology.org/2023.findings-emnlp.378/>.
- Carlos E Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik R Narasimhan. SWE-bench: Can language models resolve real-world github issues? In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=VTF8yNQm66>.
- Mike A Merrill, Alexander Glenn Shaw, Nicholas Carlini, Boxuan Li, Harsh Raj, Ivan Bercovich, Lin Shi, Jeong Yeon Shin, Thomas Walshe, E. Kelly Buchanan, Junhong Shen, Guanghao Ye, Haowei Lin, Jason Poulos, Maoyu Wang, Marianna Nezhurina, Di Lu, Orfeas Menis Mastromichalakis, Zhiwei Xu, Zizhao Chen, Yue Liu, Robert Zhang, Leon Liangyu Chen, Anurag Kashyap, Jan-Lucas Uslu, Jeffrey Li, Jianbo Wu, Minghao Yan, Song Bian, Vedang Sharma, Ke Sun, Steven Dillmann, Akshay Anand, Andrew Lanpouthakoun, Bardia Koopah, Changran Hu, Etash Kumar Guha, Gabriel H. S. Dreiman, Jiacheng Zhu, Karl Krauth, Li Zhong, Niklas Muennighoff, Robert Kwesi Amanfu, Shangyin Tan, Shreyas Pimpalgaonkar, Tushar Aggarwal, Xiangning Lin, Xin Lan, Xuandong Zhao, Yiqing Liang, Yuanli Wang, Zilong Wang, Changzhi Zhou, David Heineman, Hange Liu, Harsh Trivedi, John Yang, Junhong Lin, Manish Shetty, Michael Yang, Nabil Omi, Negin Raoof, Shanda Li, Terry Yue Zhuo, Wuwei Lin, Yiwei Dai, Yuxin Wang, Wenhao Chai, Shang Zhou, Dariush Wahdany, Ziyu She, Jiaming Hu, Zhikang Dong, Yuxuan Zhu, Sasha Cui, Ahson Saiyed, Arinbjörn Kolbeinsson, Christopher Michael Rytting, Ryan Marten, Yixin Wang, Jenia Jitsev, Alex Dimakis, Andy Konwinski, and Ludwig Schmidt. Terminal-bench: Benchmarking agents on hard, realistic tasks in command line interfaces. In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=a7Qa4CcHak>.
- Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin Lin, and Chuan Wu. Hybridflow: A flexible and efficient rlhf framework. *arXiv preprint arXiv: 2409.19256*, 2024.
- Jason Wei, Maarten Bosma, Vincent Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V Le. Finetuned language models are zero-shot learners. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=gEZrGCozdqR>.
- Zihan Wang, Kangrui Wang, Qineng Wang, Pingyue Zhang, Linjie Li, Zhengyuan Yang, Kefan Yu, Minh Nhat Nguyen, Licheng Liu, Eli Gottlieb, et al. Ragen: Understanding self-evolution in llm agents via multi-turn reinforcement learning. *arXiv preprint arXiv:2504.20073*, 2025.
- Taiwei Shi, Yiyang Wu, Linxin Song, Tianyi Zhou, and Jieyu Zhao. Efficient reinforcement finetuning via adaptive curriculum learning. *arXiv preprint arXiv:2504.05520*, 2025.
- Yanli Zhao, Andrew Gu, Rohan Varma, Liang Luo, Chien-Chin Huang, Min Xu, Less Wright, Hamid Shojanazeri, Myle Ott, Sam Shleifer, Alban Desmaison, Can Balioglu, Pritam Damania, Bernard Nguyen, Geeta Chauhan, Yuchen Hao, Aijt Mathews, and Shen Li. Pytorch fsdp: Experiences on scaling fully sharded data parallel. *Proc. VLDB Endow.*, 16(12): 3848–3860, August 2023. ISSN 2150-8097. doi: 10.14778/3611540.3611569. URL <https://doi.org/10.14778/3611540.3611569>.

Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. Pytorch: An imperative style, high-performance deep learning library. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 32. Curran Associates, Inc., 2019. URL https://proceedings.neurips.cc/paper_files/paper/2019/file/bdbca288fee7f92f2bfa9f7012727740-Paper.pdf.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with pagedattention. In *Proceedings of the ACM SIGOPS 29th Symposium on Operating Systems Principles*, 2023.

A. Ethical Considerations

We adhere to ethical principles to ensure the responsible development and application of our proposed techniques. The research conducted in the paper conforms in every respect, with the NeurIPS code of ethics guidelines. Our work focuses on enhancing the rule-based reasoning abilities of SLMs without directly involving human subjects or sensitive information. The study also acknowledges several ethical implications such as the transparency of rule-based systems, though advantageous for interpretability, raises concerns about accountability if rules are misapplied in high-stakes domains. We advocate for rigorous validation of rules against diverse datasets in avoid of human biases on manually crafted rules. We also acknowledge the potential broader impacts of our research, recognize the environmental and computational costs associated with LLM training, and strive to optimize our methods for efficiency.

B. Prompts

In this work, we use the same prompt template for each dataset for model training and evaluation.

B.1. Prompts for Dataset with Explicit Rules.

Instruction: Please answer the question based on the given rules and facts using either of [A/B/C/D] (or [True/False/Unknown]). Fill in the answer between <answer> and </answer>. Provide your step by step reasoning process between <think> and </think>.

Input:

Rules: {Rules}

Facts: {Facts}

Question: {Question}

Options: {Options} (OPTIONAL)

B.2. Prompts for Dataset with Implicit Rules.

Instruction: Please answer the question based on the given contexts using either of [A/B/C/D] (or [True/False/Unknown]). Fill in the answer between <answer> and </answer>. Provide your step by step reasoning process between <think> and </think>.

Input:

Context: {Context}

Question: {Question}

Options: {Options} (OPTIONAL)

C. Data Details

C.1. Data Sources

We list the training and evaluation data sources associated with the urls used in the paper as below. The followings are the training and validation data sources:

C.2. Dataset Curation Details

For ProntoQA, we randomly negate some of the proof questions to avoid learning the shortcut of answer "True". For ProofWriter, we randomly sample ten percent of the original source data considering the imbalance nature of the whole training data. Then we use DeepSeek-R1 to generate the reasoning process including short CoT and long CoT sequences for each QA sample. For LogiQA, we use data with the reasoning type both categorical reasoning and conjunctive reasoning that leverages the implicit

rule application and reasoning. For BigBench-Hard, we use the subset of “logical_deduction” with three, five, and seven objects with varying levels of difficulties and select BoardgameQA and ZebraPuzzles from BigBench-Extra-Hard to keep consistent with our task definition for OOD evaluation.

D. Additional Evaluation Results

AGENTGEM can Adapt to Varying Rule Complexity.

Test-time Scalability.

E. Computational Infrastructure & Hyperparameters

We list the details of the computational infrastructure and hyperparameters of training and inference used in this work in Table 3.

Table 3: Hyper-parameters of AGENTGEM-4B and AGENTGEM-8B on-policy training and inference.

Computational Infrastructure 4 × A100-SXM4-80GB GPU			
Hyperparameter	Assignment	Hyperparameter	Assignment
Base model	AGENTGEM-4B (Qwen3-4B-Base)	Base model	AGENTGEM-8B (Qwen3-8B-Base)
Training strategy	FSDP (Zhao et al., 2023)	Training strategy	FSDP (Zhao et al., 2023)
Training epochs	3	Training epochs	3
Training precision	bfloat16	Training precision	bfloat16
Mini batch size	65,536 tokens	Mini batch size	65,536 tokens
Micro batch size	16,384 tokens	Micro batch size	16,384 tokens
Max input length	1,024	Max input length	1,024
Max output length	2,048	Max output length	2,048
Max learning rate	1e − 6	Max learning rate	1e − 6
Optimizer	AdamW	Optimizer	AdamW
Adam beta weights	0.9, 0.95	Adam beta weights	0.9, 0.95
KL loss coefficient	0	KL loss coefficient	0
Rollout per sample	64	Rollout per sample	64
Clip ratio	0.2	Clip ratio	0.2
Gradient clipping	1.0	Gradient clipping	1.0
Evaluation steps	10	Evaluation steps	10
Save steps	10	Save steps	10
Training engine	PyTorch (Paszke et al., 2019)	Training engine	PyTorch (Paszke et al., 2019)
Decay factor α	0.5	Decay factor α	0.5
Base reward ϵ	0.1	Base reward ϵ	0.1
Scaling factor τ	0.8	Scaling factor τ	0.5
Rollout engine	vLLM (Kwon et al., 2023)	Rollout engine	vLLM (Kwon et al., 2023)
Rollout temperature	0.8	Rollout temperature	0.8
Rollout top-p	1	Rollout top-p	1
Random seeds	{21, 42, 84}	Random seeds	{21, 42, 84}

F. Exemplar Trajectories

We provide additional positive and negative examples (*i.e.*, failed cases) with fine-grained reasoning trajectories from the test sets of various tasks to demonstrate clear step-by-step logical reasoning patterns and well-organized thought process of AGENTGEM after training in ?? and ??.